Cagri Demir *Software Engineer · M.Sc. Student (AI/ML)*

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Summary

Software engineer and **M.Sc. student in Applied Computer Science** at **Ruhr-Universität Bochum** (focus: machine learning) with **6+ years of production iOS development** (**Swift**, **SwiftUI**) — banking apps with **250K+ users** and e-commerce apps for brands such as **Nike**. Hands-on ML ranges from electricity-demand forecasting pipelines (**Python**, **Apache Airflow**) and explainable deep-learning image classification (**PyTorch**, **Grad-CAM**) to NLP and classical ML (**scikit-learn**, **SHAP**) and **LLM integration** (Claude API). Production experience adds CI/CD, automated testing, monitoring, and secure APIs — the engineering it takes to move models from notebook to product.

Availability: immediately as a working student, up to 20 h/week during lectures (more during semester breaks), until approx. 09/2028. Student residence permit – working-student employment permitted.

ML & AI: Python, PyTorch, scikit-learn, pandas, NumPy, SHAP, Grad-CAM, Streamlit, Claude API
Data: SQL (MySQL), Apache Airflow, Apache NiFi, Elasticsearch, Tableau, Kibana
Engineering: Git/GitLab, CI/CD (Bitrise, CircleCI, Fastlane), unit & UI testing, REST APIs, Claude Code (agentic development)
Mobile (iOS): Swift, SwiftUI, UIKit, MVVM, Clean Architecture, XCTest, WidgetKit, HealthKit, Apple Pay, Firebase

Professional Experience

12/2022 – 07/2026 **Software Engineer (iOS),** *OBSS – BKT (Banka Kombëtare Tregtare), Istanbul*

- Developed **BKT Smart & BKT Kosova Mobile** — banking apps serving **250K+ users** in Albania and Kosovo — with Clean Architecture and SwiftUI; integrated and documented **REST APIs** (Swagger); implemented a secure-payment stack (JWT, SSL pinning, Keychain).
- Maintained a **99%+ crash-free user rate** (Firebase Crashlytics) through unit and UI tests (**XCTest**), **CI/CD** (Bitrise), and production monitoring (Countly), working in multinational Agile/Scrum teams in English.

10/2020 – 12/2022 **iOS Engineer,** *EPAM Systems / Emakina, Izmir*

- Delivered iOS features for **Nike**, **Under Armour ME**, and **Bloomingdale's Middle East** across the UAE, Saudi Arabia, and Kuwait; **Objective-C → Swift** migration with MVVM and SOLID principles.
- Managed **CI/CD** (CircleCI), code quality (SonarQube), and analytics tracking (Firebase, Snowplow, Google Analytics); integrated Apple Pay, Face ID, and bilingual RTL interfaces.

04/2020 – 07/2020 **Junior Data Scientist,** *Onur Enerji, Izmir*

- Built end-to-end **ML pipelines** in **Python** for electricity-consumption forecasting and demand-peak prevention with regression and ensemble models; orchestrated with **Apache Airflow**.
- Engineered features from energy and weather data (WeatherMap API); ran EDA and built dashboards (Tableau, Kibana, R); stored results in **MySQL** and **Elasticsearch**.

05/2019 – 08/2019 **Software Engineer Intern,** *Onur Enerji, Izmir*

- Built **ETL pipelines** with Apache NiFi and Java consuming EPIAS REST APIs; processed JSON electricity-market data into MySQL; supported Elasticsearch indexing for carbon.ninja.

AI / ML Projects

05/2026 **Chest X-Ray Pneumonia Detection,** *PyTorch, Python* · GitHub

- Transfer learning with **EfficientNet-B0** on the Kaggle Chest X-Ray Pneumonia dataset (5,800+ images); two-phase fine-tuning with a class-weighted loss for the 3:1 class imbalance.
- **87.7% test accuracy, ROC-AUC 0.96** on 624 held-out images; **Grad-CAM** heatmaps explain each prediction (**XAI**); interactive Streamlit demo.

04/2026 – Present **Sana — AI Nutrition Coach,** *SwiftUI + HealthKit + Claude API* · App Store

- **Photo-based meal analysis** (calories, macros, health score) with the **Claude vision API**, barcode scanning, and **LLM-powered** meal recommendations personalised with HealthKit data, allergies, and diet type; 6-language localisation; live on the App Store.
- Built solo with **agentic coding workflows (Claude Code)** across the SwiftUI codebase, API integration, testing, and localisation.

05/2026 **Tech Job Market Intelligence,** *Python, scikit-learn, SHAP, Streamlit* · GitHub

- **NLP** on job titles (TF-IDF, LDA topic modelling) across 9K+ data-job postings; salary regression comparing Ridge, Random Forest, and **Gradient Boosting**, explained with **SHAP**.
- Role segmentation with **K-Means** and PCA; three-tab Streamlit app for market exploration and salary prediction.

05/2026 **ML Job Analyzer,** *Python, pandas, scikit-learn, Streamlit* · GitHub

- **Random Forest** classifier predicting the seniority level (Entry / Mid-Senior / Director) of LinkedIn job postings from salary, work type, remote option, and title keywords; 123K-posting Kaggle dataset filtered to 28K complete records; **72% accuracy**; Streamlit predictor.

Education

10/2026 – Present **M.Sc. Applied Computer Science (Angewandte Informatik),** *Ruhr-Universität Bochum*

- Focus: machine learning · Current coursework: Deep Learning, NLP with Deep Learning, AI Ethics and Society · Expected graduation: 09/2028

08/2015 – 08/2020 **B.Sc. Software Engineering,** *Beykent University, Istanbul*

- English-medium programme; ZAB (KMK) evaluation: equivalent to a German Bachelor's degree.
- Graduation project (two semesters, grades A and A−): ARObjectDetection — iOS app (ARKit + Firebase) for real-time object recognition.

Languages

Turkish — Native
English — Fluent; working language for 6+ years
German — C1 (telc Deutsch C1 Hochschule)

Certifications

09/2026 Machine Learning Specialization, Stanford Online / DeepLearning.AI (Coursera)
09/2026 The LLM Course – Fundamentals of LLMs, Hugging Face
06/2026 Google AI Professional Certificate, Coursera